Day 4: Firewall BasicsTheory

Q1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.Steps followed

Practical: Block Spotify Using Windows Defender Firewall

Aim: To create a new outbound firewall rule in Windows Defender Firewall to block network traffic from the Spotify desktop application.

Steps to Perform

1. Press **Windows + R**, type wf.msc, and press Enter.
2. The **Windows Defender Firewall with Advanced Security** window will open.
3. Click **Outbound Rules** from the left panel.
4. Click **New Rule** from the right-side Actions panel.
5. Select **Program** and click **Next**.
6. Select **This program path** and browse to the Spotify application's executable file (Spotify.exe).
7. Click **Next** and select **Block the connection**.
8. Select all three profiles: **Domain, Private, and Public**.
9. Click **Next** and enter the rule name **Block Spotify**.
10. Click **Finish** to create the outbound firewall rule.

Verification

1. Open Windows Defender Firewall with Advanced Security.
2. Select **Outbound Rules**.
3. Find the **Block Spotify** rule and verify that it is enabled and its action is set to **Block**.
4. Open Spotify and test whether it can access online content.

Observation

The outbound firewall rule blocks network traffic from the selected Spotify executable file. Spotify may fail to load online content or connect to its servers.

Conclusion

Successfully created an outbound firewall rule in Windows Defender Firewall to block Spotify's network access. This practical demonstrates how firewall rules can control application-level network traffic and improve network security.



Q2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Practical: Configure an Inbound Firewall Rule for Google Chrome (HTTPS)
Aim: To create an inbound rule in Windows Defender Firewall that allows incoming TCP connections on port 443 (HTTPS) for Google Chrome.
Steps to Perform

1. Press **Windows + R**, type wf.msc, and press Enter.
2. Open **Windows Defender Firewall with Advanced Security**.
3. Click **Inbound Rules** from the left panel.
4. Click **New Rule** from the right-side Actions panel.
5. Select **Custom** and click **Next**.
6. Select **This program path** and browse to the Google Chrome executable file (chrome.exe).
7. Under **Protocol and Ports**, select **TCP** and set **Local Port** to **443**.
8. Keep the required IP address settings and click **Next**.
9. Select **Allow the connection** and click **Next**.
10. Select the required profiles: **Domain, Private, and Public**.
11. Enter the rule name **Allow Chrome HTTPS Inbound**.
12. Click **Finish** to create the firewall rule.

Verification

1. Open **Inbound Rules** in Windows Defender Firewall.
2. Find the rule named **Allow Chrome HTTPS Inbound**.
3. Verify that the rule is enabled and its action is **Allow**.
4. Open the rule properties and confirm:
    - Program: chrome.exe
    - Protocol: TCP
    - Local Port: 443
    - Action: Allow

Observation
The firewall rule permits inbound TCP connections on local port 443 for the Google Chrome executable, subject to other applicable firewall rules.
Conclusion
Successfully configured an inbound firewall rule for Google Chrome on TCP port 443 using Windows Defender Firewall with Advanced Security.

Q3 Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Use the 'Scope' tab when creating a new rule to specify the IP address.

Practical: Block a Specific IP Address Using Windows Defender Firewall

Aim: To block all incoming and outgoing network traffic from a specific IP address (123.45.67.89) using Windows Defender Firewall.

Steps to Perform

1. Press **Windows + R**, type wf.msc, and press **Enter**.
2. Windows Defender Firewall with Advanced Security will open.
3. Click **Inbound Rules** from the left panel.
4. Click **New Rule** from the right-side panel.
5. Select **Custom** and click **Next**.
6. Select **All programs** and click **Next**.
7. Under **Protocol and Ports**, select **Any** and click **Next**.
8. In the **Scope** section, find **Which remote IP addresses does this rule apply to?**
9. Select **These IP addresses**, click **Add**, and enter 123.45.67.89.
10. Click **OK**, then **Next**.
11. Select **Block the connection** and click **Next**.
12. Select all three profiles: **Domain, Private, and Public**.
13. Enter the rule name **Block IP 123.45.67.89 - Inbound**.
14. Click **Finish** to save the rule.
15. Repeat the same steps under **Outbound Rules** and name the rule **Block IP 123.45.67.89 - Outbound**.

Verification

1. Open Windows Defender Firewall with Advanced Security.
2. Check both **Inbound Rules** and **Outbound Rules**.
3. Verify that both rules are enabled and their action is **Block**.
4. Open each rule's **Scope** tab and confirm the remote IP address is 123.45.67.89.

Observation

Windows Defender Firewall blocks incoming and outgoing network traffic associated with the specified remote IP address.

Conclusion

Successfully configured inbound and outbound firewall rules to block communication with the IP address 123.45.67.89. This practical demonstrates how Windows Defender Firewall improves network security by restricting traffic from specific IP addresses.

Q4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Comparison Between Hardware Firewall and Software Firewall

Aim: To understand the differences between hardware and software firewalls and their uses in network security.

Key Differences

Feature	Hardware Firewall	Software Firewall
1. Installation	A physical device installed between the network and the internet.	A program installed on a computer or operating system.
2. Protection	Protects multiple devices across an entire network.	Primarily protects the individual computer on which it is installed.
3. Cost	Generally expensive and requires dedicated hardware.	Usually cheaper; Windows Defender Firewall is built into Windows.

Use Cases

1. Hardware Firewall:

Used in large offices, banks, and organizations to protect multiple computers and servers from unauthorized network access.

Example: A company uses a Cisco firewall to secure its office network.

2. Software Firewall:

Used on personal computers and laptops to control incoming and outgoing network traffic.

Example: Windows Defender Firewall blocks unauthorized applications from accessing the network.